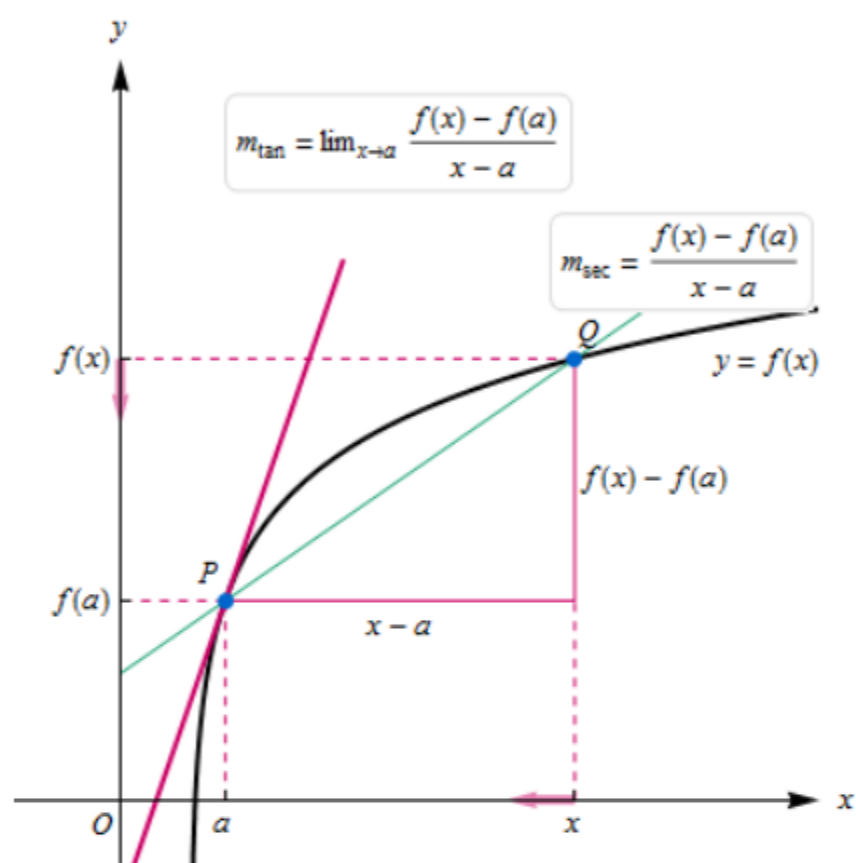


Problem 1

Problem 1

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Problem 1 Consider the following two figures depicting the same graph $y =$



Problem 1

- (a) In Figure 1 on the left, what are the coordinates of P and Q ? In Figure 2 on the right, what are the expressions for the coordinates of P and Q ?

Explanation. Figure 1 on the left: $P(a, f(a))$ and $Q(x, f(x))$

Figure 2 on the right: $P(a, f(a))$ and $Q((a + h), f(a + h))$

- (b) Express the slope of the secant line through P and Q in terms of the above coordinates for each Figure 1 and Figure 2.

Explanation. Figure 1: $\frac{f(x) - f(a)}{x - a}$ Figure 2: $\frac{f(a + h) - f(a)}{h}$

- (c) Express the slope of the tangent line at the point P in terms of the above coordinates for each Figure 1 and Figure 2.

Explanation. Figure 1: $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ Figure 2: $\lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$

- (d) What is the difference?

Explanation. Just the variable we use, x or h where $x = a + h \iff h = x - a$

- (e) For each of the two graphs, which lines are the secant lines?

Explanation. The green line is the secant line in each of the two graphs.

- (f) For each of the two graphs, which lines are the tangent lines?

Explanation. The red line is the tangent line in each of the two graphs.